



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO JAI SRI RAM AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

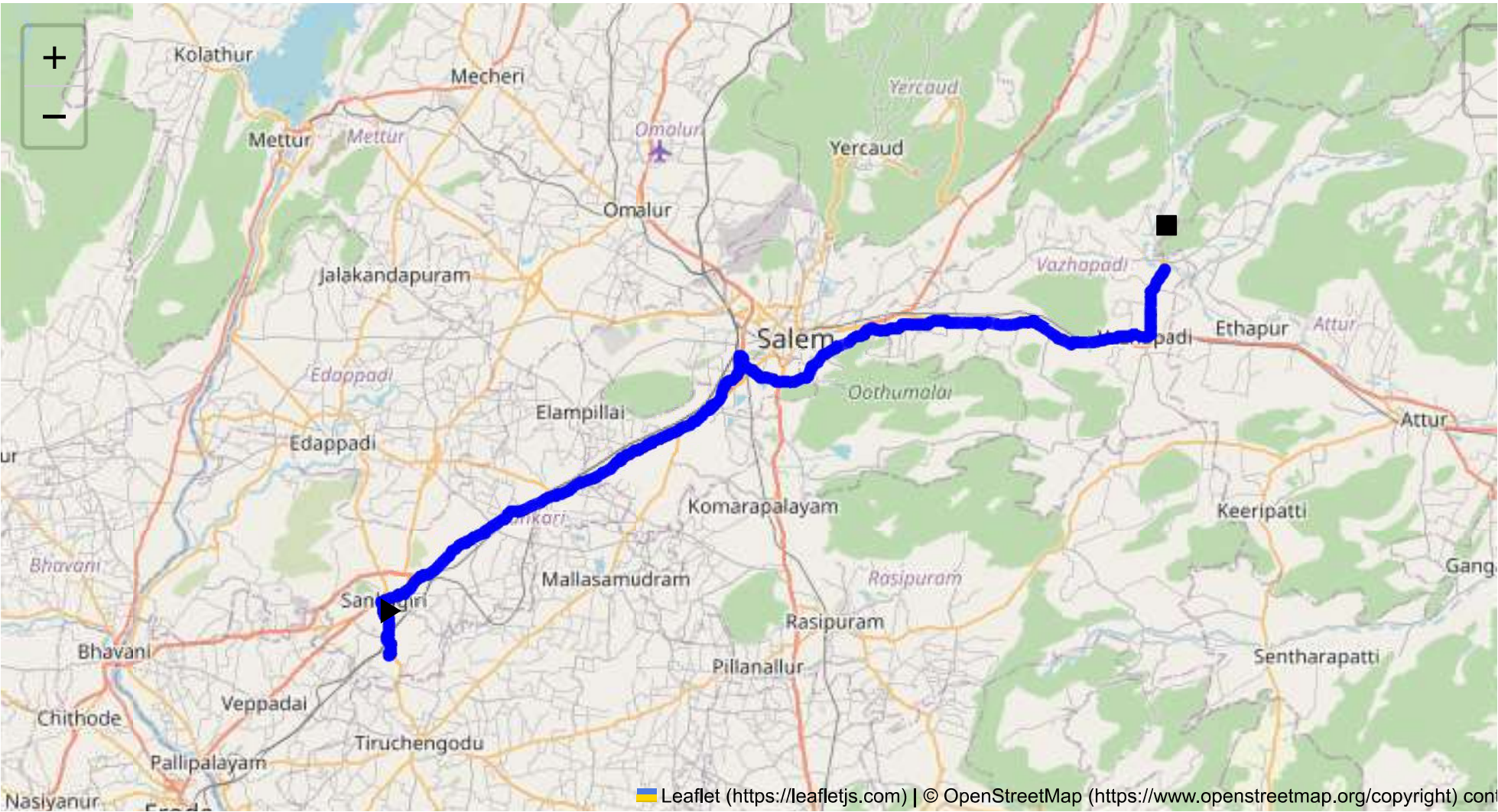
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 78.68 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (11.701582, 78.414898)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

- The route from Sangagiri, Tamil Nadu, to Belur, Tamil Nadu covers approximately 78.68 kilometers.

- It predominantly follows state highways and crosses through towns such as Tiruchengode and Kumarapalayam.
- The terrain is mostly flat with agricultural and small town areas.

2. Typical Weather Conditions and Hazards

- Tamil Nadu generally experiences a tropical climate, with hot summers and a monsoon season from October to December.
- Weather hazards include heavy monsoon rains causing potential flooding and slippery roads.
- During the summer months, high temperatures can affect vehicle operation and cause fatigue.

3. Traffic Patterns

- Peak traffic hours are typically between 8-10 AM and 6-8 PM due to local commute and market timings.
- Towns such as Tiruchengode experience higher congestion due to market activities.
- Weekends might have lighter traffic, except for local festivities or events.

4. Assessment of Road Quality and Infrastructure

- State highways in Tamil Nadu generally have moderate quality, but there may be occasional potholes and uneven surfaces.
- Infrastructure for trucks includes basic amenities with limited rest stations or truck stops.
- Some rural sections might lack proper signage and lighting.

5. Alternative Routes for Emergencies

- An alternative route through NH544 Salem-Coimbatore could be considered for avoiding local congestion or roadblocks.
- Secondary roads may be available but could be narrower and less maintained.

6. Local Regulations on Hazardous Material Transport

- Transport of hazardous materials is regulated under state laws following ADR guidelines.
- Restrictions may apply in heavily populated or sensitive areas, requiring special permits or alternative timings for transport.

7. Historical Incidents

- Reports of accidents involving heavy vehicles suggest caution around market areas in towns due to high foot traffic.
- Historically, few incidents involving hazardous materials, but standard caution advised when crossing populated zones.

8. Environmental Considerations

- The route passes through agricultural zones sensitive to chemical spills.
- Rivers and water bodies are present in the region and should be protected from contamination.

9. Communication Coverage

- While major town areas have adequate mobile and internet coverage, rural stretches may experience signal loss.
- Known dead zones for communication can typically be found in remote agricultural segments.

10. Estimated Emergency Response Times

- Urban areas like Tiruchengode can have emergency response times of approximately 15-30 minutes.
- Rural stretches may experience delays, with response times extending beyond 45 minutes.

11. Overall Risk Assessment Summary

- The journey has a moderate level of risk due to weather conditions, road quality, and traffic patterns.
- Critical risk factors include weather impacts, congestion in towns, and inadequate rest stops.
- Effective risk management includes planning travel times to avoid peak traffic, ensuring vehicle readiness for weather conditions, and staying informed of local transport regulations.

By addressing these factors, truck drivers can navigate the route safely and efficiently, minimizing risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.61736, 78.10588	30 KM/Hr
12	Turn	Medium	11.61753, 78.10608	30 KM/Hr
13	Turn	Medium	11.64172, 78.11817	30 KM/Hr
14	Turn	Medium	11.64216, 78.11875	30 KM/Hr
15	Turn	Medium	11.64195, 78.12129	30 KM/Hr
16	Turn	Medium	11.64162, 78.12160	30 KM/Hr
17	Turn	Medium	11.64120, 78.12168	30 KM/Hr
18	Turn	Medium	11.63948, 78.12081	30 KM/Hr
19	Turn	High	11.65458, 78.37988	15 KM/Hr
20	Turn	Medium	11.65484, 78.38000	30 KM/Hr
21	Turn	Medium	11.65602, 78.38711	30 KM/Hr
22	Turn	Medium	11.65631, 78.38769	30 KM/Hr
23	Turn	High	11.65716, 78.40570	15 KM/Hr
24	Turn	Medium	11.65743, 78.40584	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
13	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
14	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
17	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
21	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
22	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
24	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
25	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
26	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
27	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
29	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
30	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
31	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
32	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
37	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
39	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
40	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
41	hospital	Sri Bala Hospital	11.6566667, 78.4010889	30 km/h	Medium
42	clinic	Sibi Physiotherapy Clinic	11.6561305, 78.3994263	30 km/h	Medium
43	hospital	Saraswathi Hospital, Valapady	11.6563505, 78.4009062	30 km/h	Medium
44	hospital	Devaki Hospital	11.6564189, 78.402376	30 km/h	Medium
45	hospital	Sri Udaya Hospital	11.6560244, 78.4040862	30 km/h	Medium
46	hospital	Government Hospital, Valapady	11.65706, 78.403773	30 km/h	Medium
47	hospital	Deepam Hospital	11.6576703, 78.4065155	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
18	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
19	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
20	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
23	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
33	school	St.joseph's matric hr.sec school	11.6419382, 78.1709055	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
35	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
36	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
38	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713

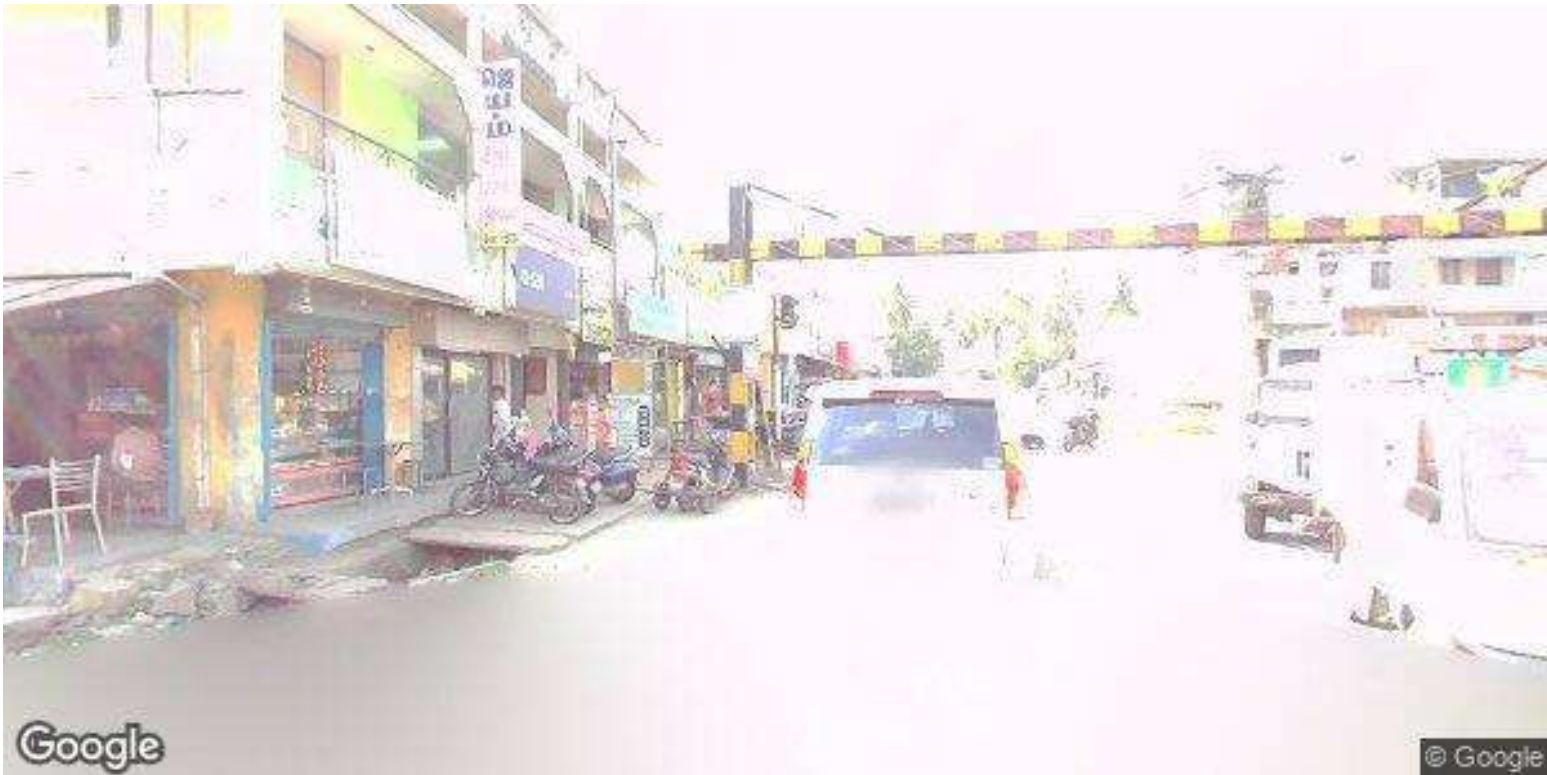


Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64172, 78.11817



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65458, 78.37988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65484, 78.38000



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65602, 78.38711



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65631, 78.38769



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65716, 78.40570



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65743, 78.40584